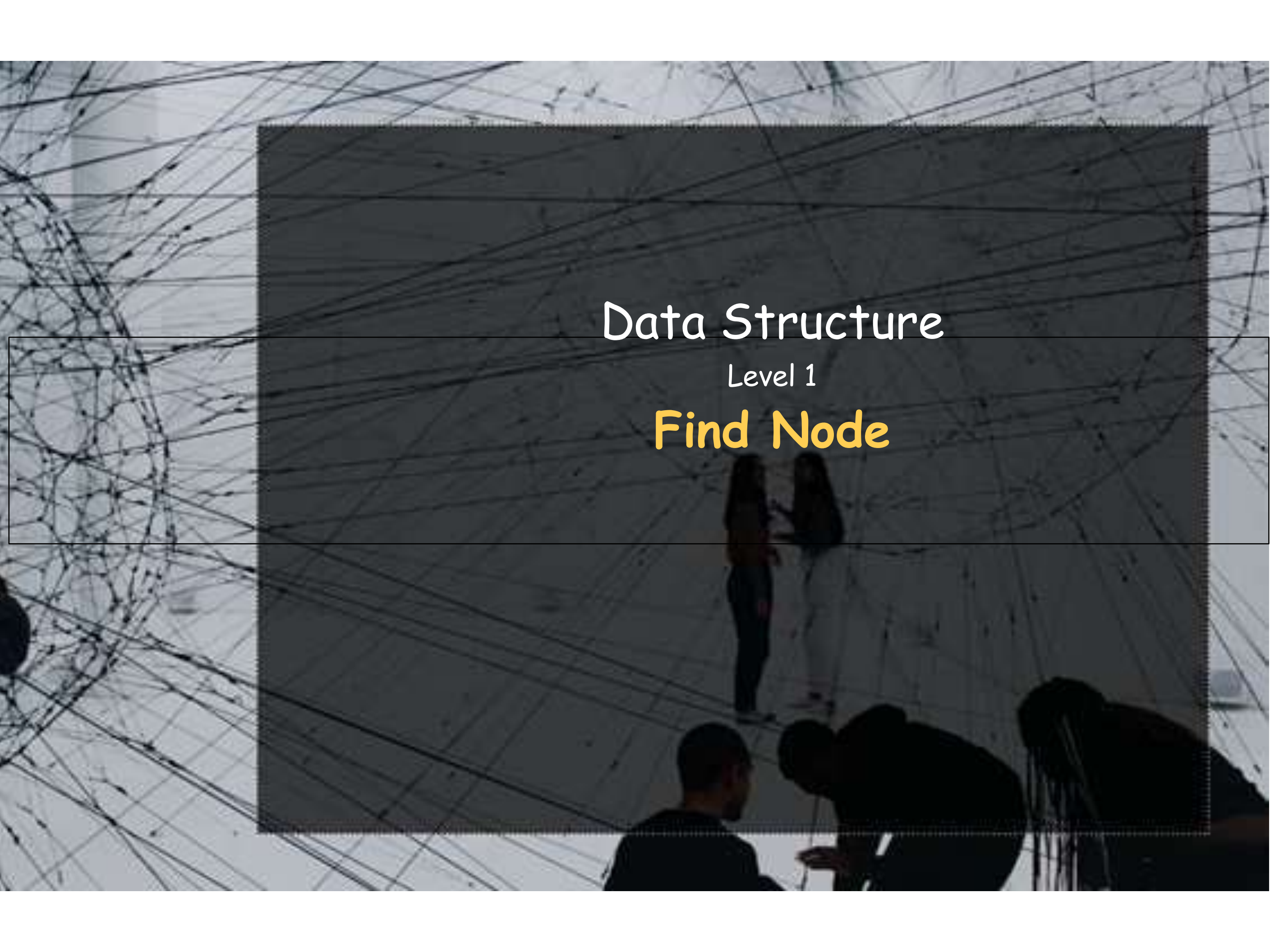


Data Structure

Level 1

Doubly Linked List Operation



Data Structure

Level 1

Find Node

Definition

A Doubly Linked List is a data structure where each node contains:

- A value
- A pointer to the next node
- A pointer to the previous node

This allows traversal in both directions (forward and backward).

Explanation

◆ Node Structure

- Each node contains:
 - value → stores the data
 - next → pointer to the next node
 - prev → pointer to the previous node

💡 Example:

NULL <- 1 <-> 2 <-> 3 -> NULL

□ Procedures (Insert at Beginning)

① InsertAtBeginning

□ Adds a new node at the start of the list

□ Steps:

① Create a new node

② Set its value

③ Point its next to current head

④ Set its prev to NULL

⑤ Update old head's prev to new node

⑥ Move head to new node

→ □ Result: New node becomes the first element

⊙ Warning: Always check if head != NULL before accessing head->prev

2 PrintNodeDetails

📖 Displays:

- ❑ Previous node value
- ❑ Current node value
- ❑ Next node value

💡 Example output:

NULL <- 1 -> 2

1 <- 2 -> 3

2 <- 3 -> NULL

3 PrintListDetails

- ▢ Traverses the list node by node and prints detailed connections

- ▢ Uses:

- ▢ Calls PrintNodeDetails for each node

4 PrintList

📖 Prints the full list in one line

💡 Example:

```
NULL <- 1 -> <- 2 -> <- 3 -> NULL
```

5 Find Function

🔍 Definition

Searches for a node with a specific value

❑ Logic:

- ❑ Start from head
- ❑ Traverse forward
- ❑ Compare each node's value
- ❑ Return the node if found
- ❑ Return NULL if not found

💡 Example:

Find(head, 2) → returns node with value 2

→ ❑ Result:

If found → pointer to node

If not found → NULL

⚖️ Comparison

□ Insert At Beginning

✓□ Time Complexity: $O(1)$

✓□ No traversal needed

✓□ Direct head update

□ Find Node

✓□ Time Complexity: $O(n)$

✓□ Requires traversal

✓□ Searches node by node

Note

The list is built in reverse order because insertion is always at the beginning

Final list becomes:

1 -> 2 -> 3 -> 4 -> 5

⊘ Warning

If you do not handle the single-node case, the program may crash.
Always update Prev->next before deleting the last node.

✦✦ Summary

- ✓ Doubly linked list allows two-way traversal
- ✓ Each node has next and prev pointers
- ✓ The Find function searches sequentially
- ✓ If found → returns node
- ✓ If not → returns NULL

Thank you 😊